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Key Points:
- Rapid uplift of the Southwest Tianshan have been initiated at approximately 6.7 Ma
- Magnetic fabrics identify the influence of the regional stress field
- The Xiyu Formation is diachronous

Supporting Information:
- Table S1–S3 Captions
- Table S1
- Table S2
- Table S3

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Neogene magnetostratigraphy and rock magnetic study of the Kashi Depression, NW China: Implications to neotectonics in the SW Tianshan Mountains

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Abstract The southwest Tianshan Mountains of China are bordered by the Tarim foreland and comprise an actively deforming segment of the India-Asia collisional system. We report a detailed magnetostratigraphic study of the Dashankou section in the Kashi Depression of the Tarim Basin to improve the understanding of the history of sedimentation, denudation, and mountain building in this region. The preferred correlation of the succession with the geomagnetic polarity timescale defines a depositional history between 12.4 and 3.0 Ma with a substantial increase in sedimentation rates identified at ~6.7 Ma corresponding to a pulse of rapid uplift in the southwest Tianshan Mountains. Although climatic changes may have modulated the record during Neogene times, they do not appear to have had an important influence on sediment accumulation rates between 7.0 and 2.6 Ma. Magnetic fabrics identify the influence of a regional stress field imparted by ongoing India-Asia collision in the lower part of the succession contrasting with predominantly sedimentary fabrics in the higher part of the succession. A major clastic influx with a maximum age estimate of ~3.6 Ma comprises the Xiyu conglomerates, and integration with other magnetostratigraphic investigations around the Tianshan demonstrates unambiguously that depositional onset of this coarse clastic episode is diachronous. Hence, the Xiyu Formation cannot be considered as a chronostratigraphic marker related to any specific tectonic or climatic event.

1. Introduction

The geologic record in the Tianshan Range attests to a complex history of Paleozoic subduction-related processes [*Burtman*, 1975; *Windley et al.*, 1990; *Xiao et al.*, 2004] and a later phase of Cenozoic reactivation induced by continental collision [*Tapponnier and Molnar*, 1977, 1979]. It is estimated that ~40% of the total India-Asia convergence has been absorbed within the Tianshan [*Abdrakhmatov et al.*, 1996; *Reigber et al.*, 2001; *Wang et al.*, 2001], and as the most active intracontinental mountain belt in the world, this orogen provides an outstanding natural laboratory for studying Cenozoic intracontinental deformation [*Burchfiel et al.*, 1999; *Sobel et al.*, 2006; *Sun et al.*, 2004].

The Tarim Basin is a foreland basin to the Tianshan where the western sector comprises the ~160 km long by 70 km wide Kashi Depression. This is overthrust by the North Pamir Thrust Fault (NPTF) in the south and by the South Tianshan Thrust Fault and the Kalpin Uplift in the north. It contains more than 6 km of Neogene and Quaternary sediments [*Heermance et al.*, 2007] providing a key opportunity for examining relationships between sedimentation and tectonic deformation.

During the past several decades, extensive research has been conducted to constrain the timing of tectonic deformation, uplift, and associated deposition linked to the Cenozoic history of Tianshan orogenesis [*Allen et al.*, 1999; *Avouac et al.*, 1993; *Bullen et al.*, 2001; *Burchfiel et al.*, 1999; *Chen et al.*, 2002; *Heermance et al.*, 2007; *Huang et al.*, 2006; *Molnar and Tapponnier*, 1975; *Tapponnier and Molnar*, 1979; *Windley et al.*, 1990; *Yin et al.*, 1998; *Zhang*, 2004]. This multidisciplinary research has included investigations of structural geology, sedimentology, and $^{40}Ar/^{39}Ar$ thermochronology [*Yin et al.*, 1998], flexural modeling linked to sedimentological

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constraints [*Yang and Liu*, 2002], apatite fission track studies [*Dumitru et al.*, 2001; *Hendrix et al.*, 1994; *Sobel and Dumitru*, 1997], and analysis of contemporary thrusting and folding [*Avouac et al.*, 1993]. The results have demonstrated that the uplift chronology varies both from east to west and north to south across the Tianshan orogen with regional variations constrained within the interval from late Oligocene to middle to late Miocene times [*Abdrakhmatov et al.*, 1996; *Amidon and Hynek*, 2010; *Bullen et al.*, 2001; *Hendrix et al.*, 1994]; a few workers have also suggested an uplift history continuing into the late Quaternary [*Burchfiel et al.*, 1999; *Fu et al.*, 2003]. However, all researchers recognize that the most important phase of uplift occurred during the middle to late Cenozoic [*Lei et al.*, 2013; *Tang et al.*, 2012].

Thick Cenozoic terrigenous deposits derived mostly from the uplifting mountain range are well preserved and exposed in the Kashi Depression [*Charreau et al.*, 2009a; *Chen et al.*, 2002; *Hendrix et al.*, 1994; *Yang and Liu*, 2002; *Huang et al.*, 2010] where the deposits divide broadly into light-colored fluviolacustrine or aeolian sediments, conglomerates, and red beds [*Li*, 1984; *Yin et al.*, 1998]. These facies contrasts are interpreted to reflect the influences of episodic uplift and variable elevations within the Tianshan Range together with climatic changes [*Li et al.*, 1979; *Yin et al.*, 1998]. In order to decipher the tectonic and climatic significance of the sedimentary record, reliable chronological control is essential. However, continental deposits typically lack diagnostic fossil floras and faunas, and when there is a paucity of datable volcanic horizons, magnetostratigraphy becomes the key tool for calibrating terrestrial deposition [*Chen et al.*, 2002; *Huang et al.*, 2006].

This technique has already been applied by a number of research groups to constrain the history of Cenozoic uplift in the Tianshan region [*Charreau et al.*, 2005, 2006, 2009a; *Chen et al.*, 2002; *Huang et al.*, 2006; *Ji et al.*, 2008; *Sun et al.*, 2004, 2005; *Yang et al.*, 2015]. These studies have recognized several pulses of sedimentation at ~24 Ma, ~17–16 Ma, ~11 Ma, ~7 Ma, and ~4 Ma interpreted to reflect multiple stages to the orogenic uplift [*Bullen et al.*, 2001; *Charreau et al.*, 2005, 2006; *Huang et al.*, 2006; *Ji et al.*, 2008; *Lu et al.*, 2010; *Sun et al.*, 2004, 2009]. The expanding number of magnetostratigraphic results has also demonstrated a diachronous uplift in the Tianshan, while interregional comparisons have highlighted chronological problems particularly concerning the timing of coarse clastic input at higher levels of the succession [*Yang et al.*, 2015].

This paper reports a new paleomagnetic study from the Mio-Pliocene foreland basin succession in the southwestern Tianshan. The studied localities are in the vicinity of the town of Jiashi where this basinal sequence is composed of ~9 km of Cenozoic sediments [*Chen et al.*, 2002; *Heermance et al.*, 2007] with important potential for documenting the tectonic and erosional histories of the southwestern Tianshan [*Jia et al.*, 2004; *Zhou and Chen*, 1990]. These rocks are now folded into a suite of elongate open anticlines and synclines thus permitting application of the paleomagnetic fold test. In the event the paleomagnetic investigation yields a detailed chronological control for the thick Cenozoic deposits in this region and provides an expanded basis for discussing the uplift history of the Tianshan Mountains since the mid-Miocene. As part of this study, we also include a new basal age estimate for the Xiyu Formation and discuss the tectonic and climatic significance of this coarse clastic facies.

2. Geological Setting, Paleomagnetic Sampling, and Methods

The southern Tianshan-northern Tarim is an outstanding example of active intracontinental thrusting and coupled sedimentation where structures and sediment thickness variations in the foreland basin reflect a strong control by deformation style imparted by the adjacent thrust system [*Allen et al.*, 1999] (Figure 1). The Neogene strata deposited within the Kashi Depression are comprehensively reviewed in a range of papers in both Chinese and English which provide the basis for this study [*Bureau of Geology and Mineral Resources of Xinjiang Autonomous Region (BGMRXUAR)*, 1993; *Chen et al.*, 2002, 2007; *Heermance et al.*, 2007, 2008; *Huang et al.*, 2009; *Jia et al.*, 2004; *Scharer et al.*, 2004; *Yin et al.*, 1998; *Zhou*, 2001]. The present investigation is concerned with that part of the succession formed during the Late Cenozoic era exposed in the Dashankou section (Figure 2).

The Neogene sequence in this section is entirely terrestrial in origin and comprises, in ascending order, the Wuqia Group, the Atushi Formation, and the largely conglomeratic Xiyu Formation [*BGMRXUAR*, 1993; *Heermance et al.*, 2007, 2008; *Huang et al.*, 2009; *Jia et al.*, 2004]. The higher part of the Wuqia Group is composed of the Pakabulaka Formation consisting of thick-bedded and massive sandstones with red brown to dark mudstones interbedded with sandy mudstones. These contain foraminifers, ostracods, chareae, and sporopollen [*Jia et al.*, 2004; *Zheng et al.*, 1999]. The formation is locally gypsiferous and is interpreted to

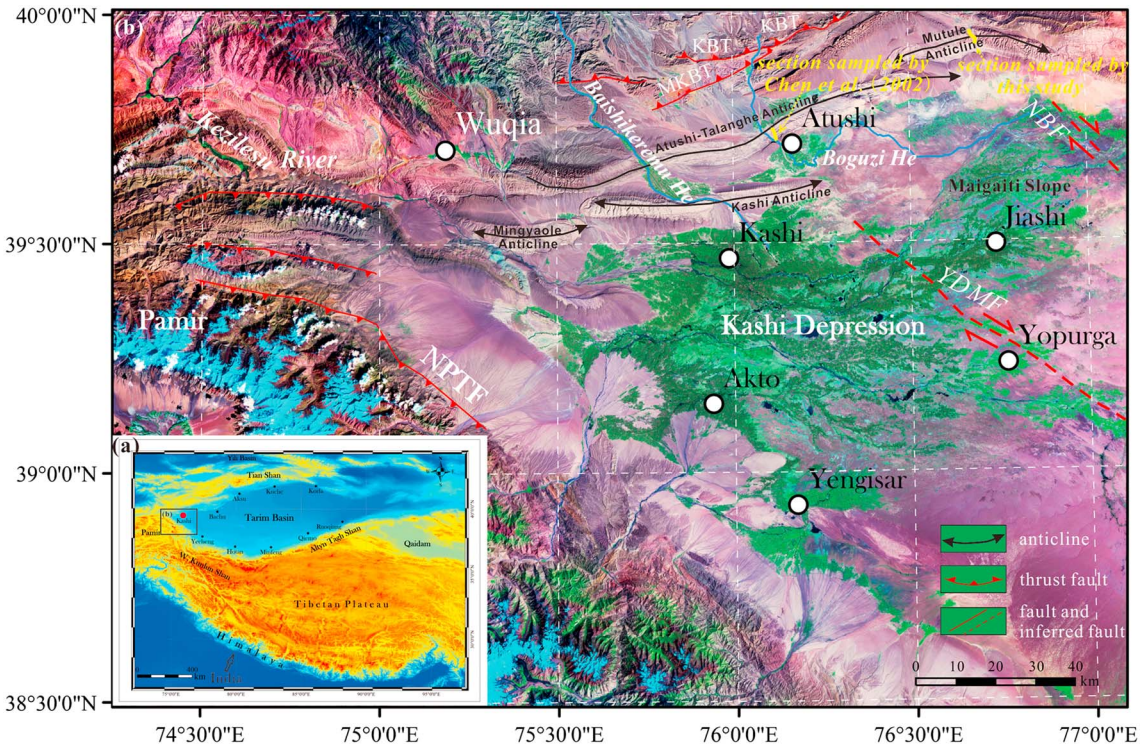


Figure 1. (a) Topographic map of the Tibetan Plateau and its hinterland in central Asia. (b) Landsat thematic mapper image of the Kashi Depression showing neotectonic features and locations of the different magnetostratigraphic studies. NBF, NPTF, and YDMF represent the North Bashituopu Fault, the North Pamir Thrust Fault, and the Yangdaman Fault, respectively.

be the product of a low-gradient, meandering fluvial system which began aggrading at ~13.7 Ma [*Heermance et al.*, 2007]. The Atushi Formation consists of yellow-brown sandstone, mudstone, gypsum (as thin interbeds of $<$10 cm thickness), and rare pebble conglomerate. These strata are similar to the Pakabulake Formation but are distinguished by their lighter color. The age of the Atushi Formation is assigned to the Pliocene from magnetostratigraphic data [*Chen et al.*, 2002] and the presence of ostracoda in the Boguzihe section [*Jia et al.*, 2004]. At the top of the Atushi there is a rapid transition into the rudaceous Xiyu Formation which, with the exception of minor interbeds of sandstone and mudstone near the base, is characterized by massive, thick-

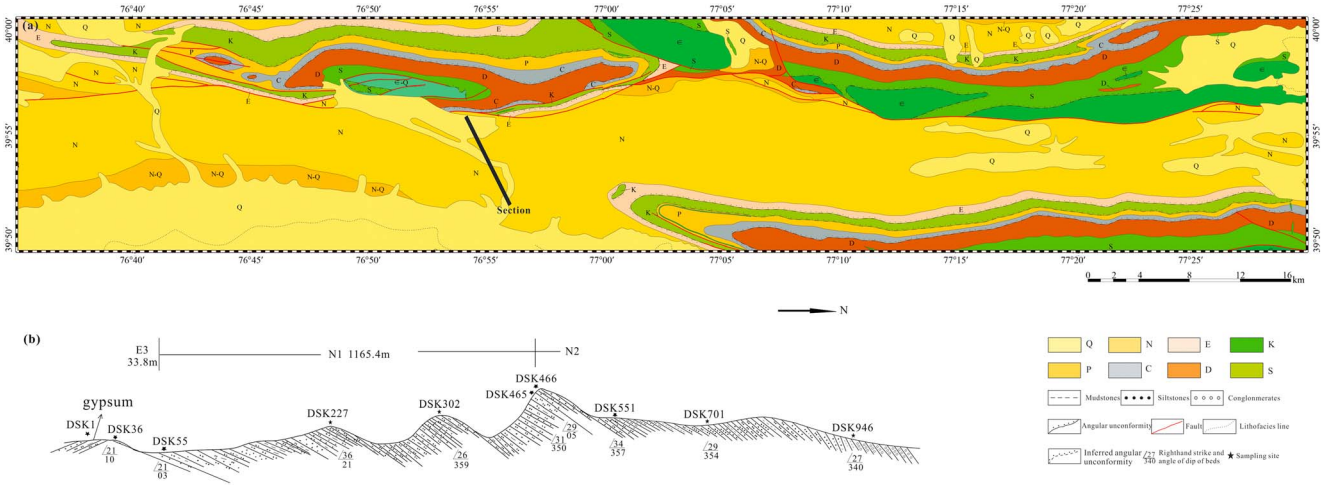


Figure 2. (a) Simplified regional geologic map of the Dashankou area showing Neogene stratigraphy and magnetostratigraphic sampling locations of this study; (b) cross section of the sedimentary succession exposed in the Dashankou area showing major stratified units and the distribution of sampling sites.

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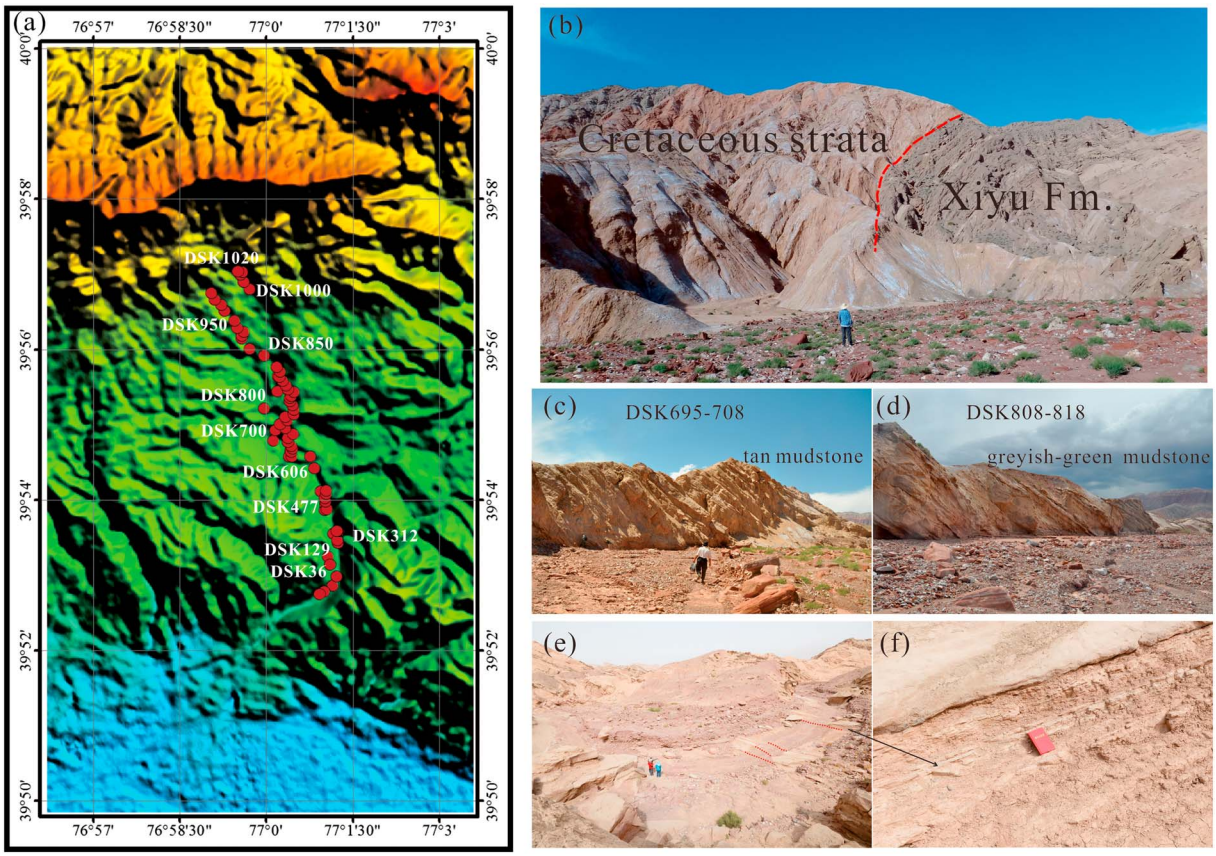


Figure 3. (a) Topographic map of the Dashankou section with sampling sites; (b) photographs showing the Xiyu conglomerate formation in the upper Dashankou section, and the Cretaceous strata overthrusting the Xiyu Formations; (c–f) photographs showing representative lithologies and facies of the section (Figures 3c and 3d), and the section cut by the river and not by neotectonic deformation (Figures 3e and 3f).

bedded, dark gray pebble to boulder conglomerates. The Xiyu Formation is widely distributed along the margins of the Tianshan, and previous paleomagnetic results in the Kashi basin have suggested that it is likely to be diachronous [*Chen et al.*, 2002; *Heermance et al.*, 2007].

The magnetostratigraphic analysis of the Dashankou section is derived from a sampled section spanning a thickness of 3187 m (Figure 3). This section commences in the northern limb of an anticline where 985 horizons were selected for paleomagnetic sampling and at least two cores were drilled from each horizon using a portable petrol-powered drill; these were then oriented with magnetic compass and, where possible, by sun compass. The sampling intervals are typically 2–5 m between successive horizons; occasionally increased by up to ~ 10 m where the sequence is dominated by conglomerate or coarse sandstone facies. In addition, eight sites (FT1–FT8) comprising 8 to 10 drill core samples per site were collected from the southern limb of the anticline to permit a paleomagnetic fold test. A local declination of 3° determined from the present International Geomagnetic Reference Field in the sampling area has been used to correct drill core azimuths and bedding attitudes.

Core specimens 2.0 cm in length and 2.5 cm in diameter were cut from field samples in the laboratory, and selected end core material was used for rock magnetic investigation. Following evaluation of rock magnetic behaviors resolved from pilot samples, all specimens were subject to progressive thermal demagnetization employing a TD-48 thermal demagnetizer. Measurements were performed using a 2G-760R U-channel system sited in a magnetically shielded room with the ambient magnetic field minimized to $<$300 nT in the Paleomagnetism and Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences. Samples were typically demagnetized in steps of 50 or 100°C up to 500°C, subsequently reduced to steps of 30, 20, or 10°C as the maximum unblocking temperatures of the remanence carriers were approached. Demagnetization results were evaluated by stereographic projections and orthogonal diagrams

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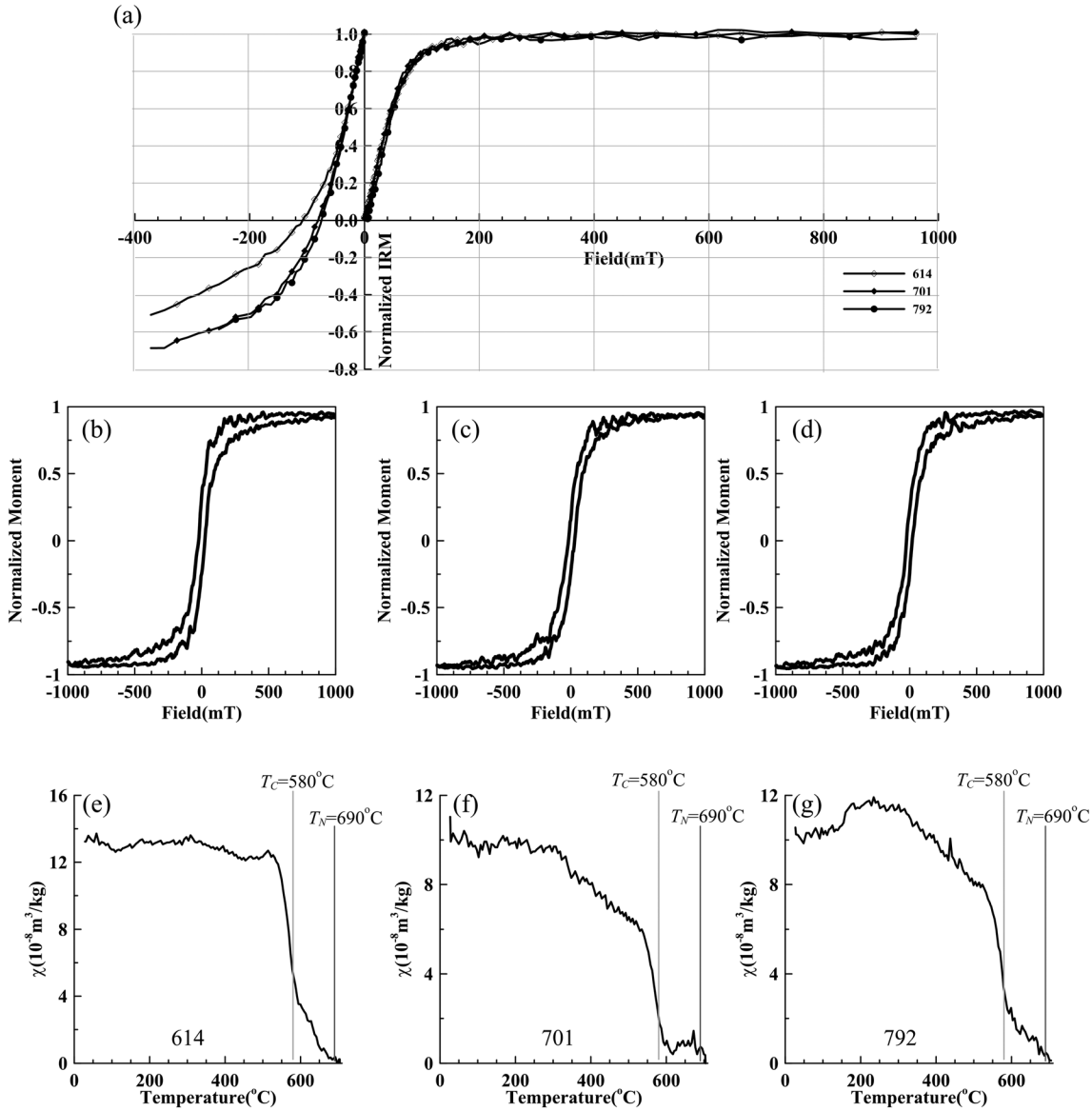


Figure 4. Rock magnetic behaviors of typical pilot samples from the Dashankou section, Kashi Depression: (a) example of acquisition behavior and backfield demagnetization of isothermal remanent magnetization; (b–d) hysteresis loop; (e–g) temperature dependence of magnetic susceptibility (χ-T curves).

[*Zijderveld*, 1967] with the latter used to resolve component structures employing principal component analysis [*Kirschvink*, 1980]. The resulting paleomagnetic directions were analyzed using standard Fisher statistics [*Fisher*, 1953] to determine interval mean directions; the associated statistical parameters were calculated using paleomagnetic software packages of R. Enkin (unpublished).

3. Paleomagnetic Results

A typical specimen shows quasi-saturation behavior by 0.2–0.6 T with remanent coercivity ranging from 75 to 80 mT (Figure 4a) indicating a spectrum dominated by a fairly low coercivity magnetic mineral. Hysteresis loops are typically wasp waisted (Figures 4b–4d) illustrating a ferrimagnetism (in a broad sense) dominated either by both soft and hard coercivity magnetic minerals or by a single ferromagnet with a broad grain size distribution [*Roberts et al.*, 1995]. However, a common characteristic in all measured samples is for the heating curves to show a noticeable decrease in magnetic susceptibility near the Curie/Néel temperatures of both magnetite and hematite (Figures 4e–4g). The combined evidence from thermomagnetic analysis and the

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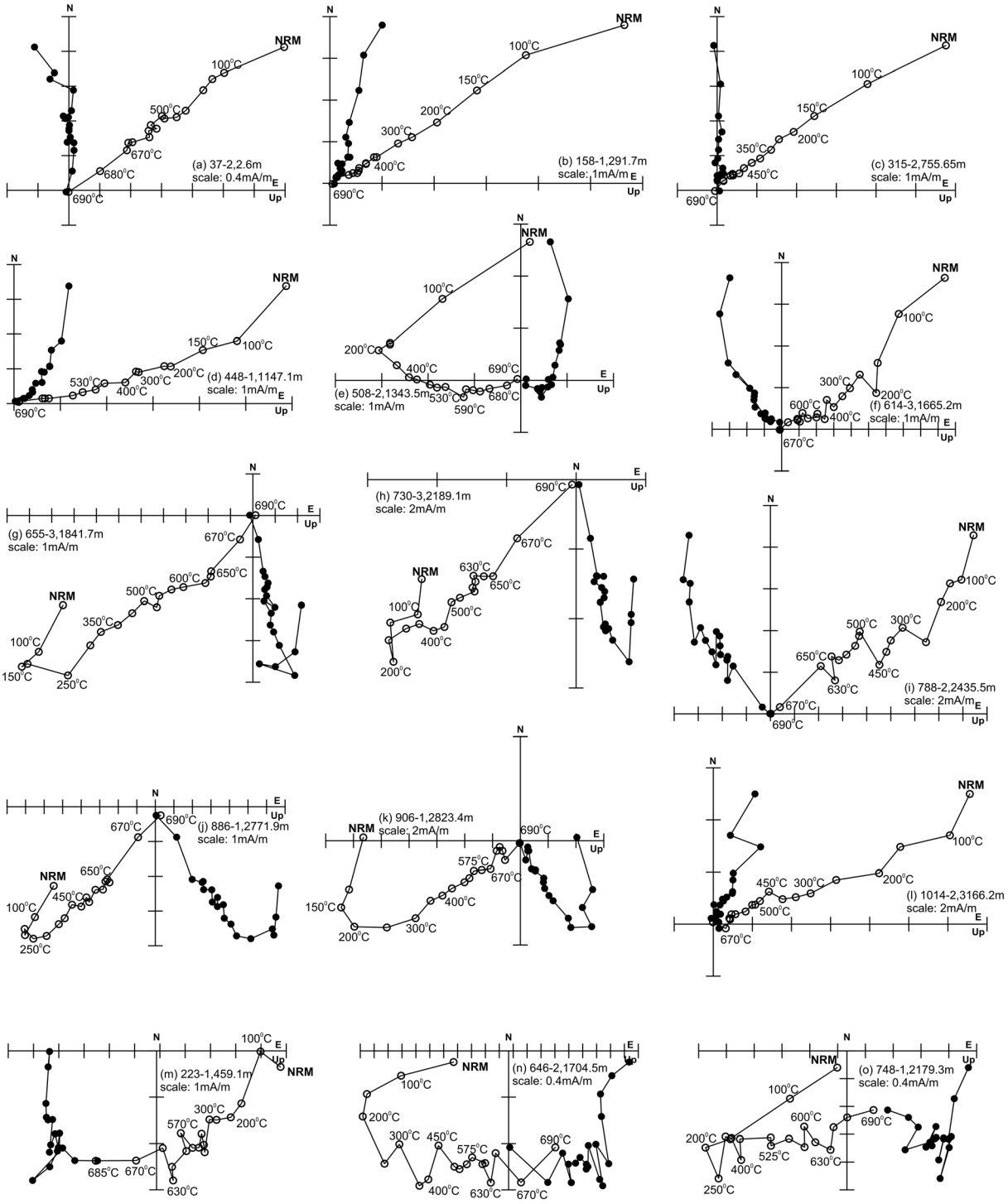


Figure 5. Orthogonal (Zijderveld) vector plots of representative specimens from different levels in the Dashankou section. Directions are plotted in situ; solid and open circles represent vector endpoints projected onto horizontal and vertical planes, respectively.

wasp-waisted loops (Figures 4b–4d) therefore indicates that both magnetite and hematite are generally present as remanence carriers.

In most specimens progressive demagnetization defines relatively straightforward trajectories moving toward the origin of the orthogonal plots following removal of a viscous overprint at temperatures generally below

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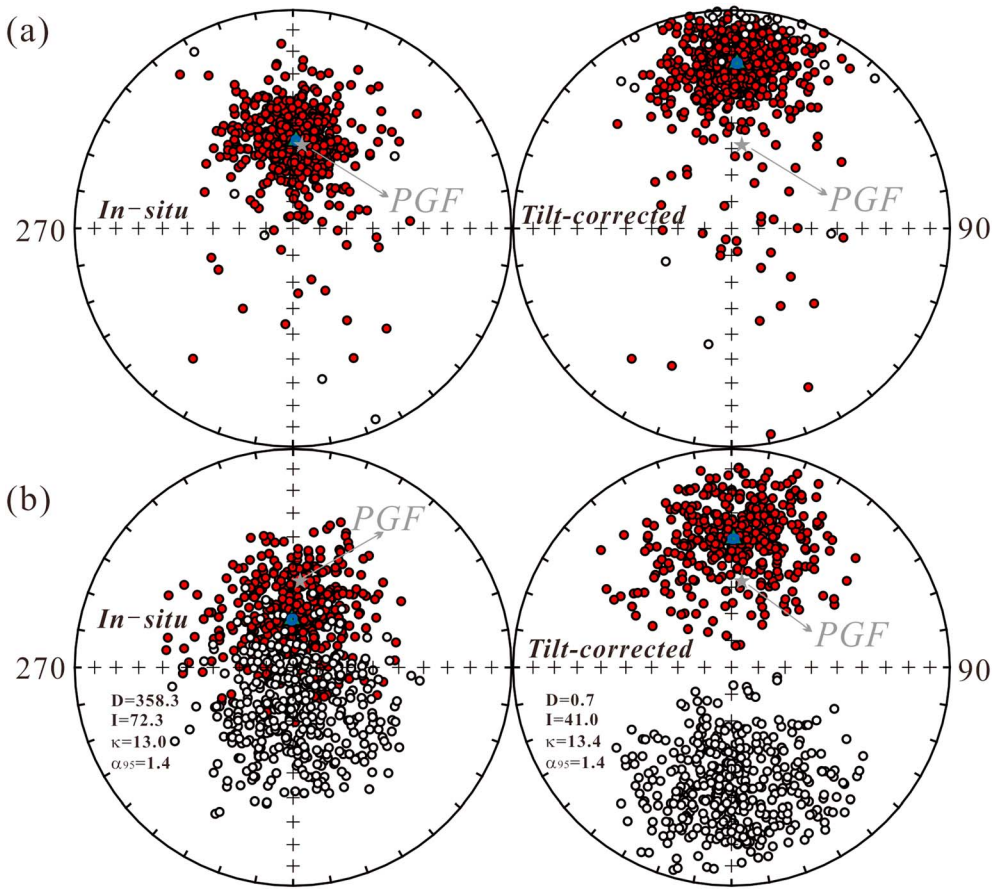


Figure 6. Equal-area projections of remanence directions from sampled horizons and overall mean directions (triangles) with 95% confidence ellipses before and after tile correction: (a) the directions of low unblocking temperature remanent magnetization; (b) the directions of high-temperature characteristic remanent magnetizations. Solid/open symbols represent downward/upward inclinations; PGF indicates the direction of present geomagnetic field.

300°C (Figure 5). The low-temperature component is predominantly of normal polarity and clusters around the present geomagnetic field direction. The group mean direction of this component with declination $D = 3.7°$ and inclination $I = 59.8°$ correlates closely with the predicted geocentric axial dipole direction of $D = 2.2°$, $I = 57.3°$; the significant deterioration in directional grouping following unfolding further supports a Recent origin (Figure 6).

Figure 5 showing Zijderveld plots of representative specimens illustrating the typical dual component behaviors with components fully subtracted mostly over temperature steps between 200–300°C and 670–690°C, respectively. The characteristic remanent magnetizations (ChRMs) are of dual polarity. By indicating that hematite and more rarely magnetite are the dominant carriers of the ChRM, the demagnetization characteristics conform to the results of thermomagnetic analysis (Figure 5) and to the terrestrial origin and variably oxidized nature of the bulk of the samples. Following demagnetization, 985 sampled horizons yield well-defined ChRMs with 786 component directions clustering within a circle of 40° radius around the mean; the residue are plausibly attributable to geomagnetic transitions or excursions. The ChRM directions of the majority group yield an overall mean of $D = 358.3°$, $I = 72.3°$ ($k = 13.0$, $\alpha_{95} = 1.4°$) before and $D = 0.7°$, $I = 41.0°$ ($k = 13.4$, $\alpha_{95} = 1.4°$) after tilt adjustment with no evident improvement in directional grouping following complete unfolding (Figure 6b). However, analysis using the Watson and Enkin [*Watson and Enkin*, 1993] test identifies an optimal concentration at 91.4% unfolding with the 95% uncertainties ranging from 88.9 to 94.0% unfolding indicating that the ChRM has a prefolding origin. The reversal test [*McFadden and McElhinny*, 1990] is indeterminate for the whole sampled section, but when the ChRM directions are grouped stratigraphically into two subsections, the reversal test is positive: the upper part of the Dashankou section yields an angular difference of 3.8° between tilt-adjusted interval mean directions of each polarity which is less than

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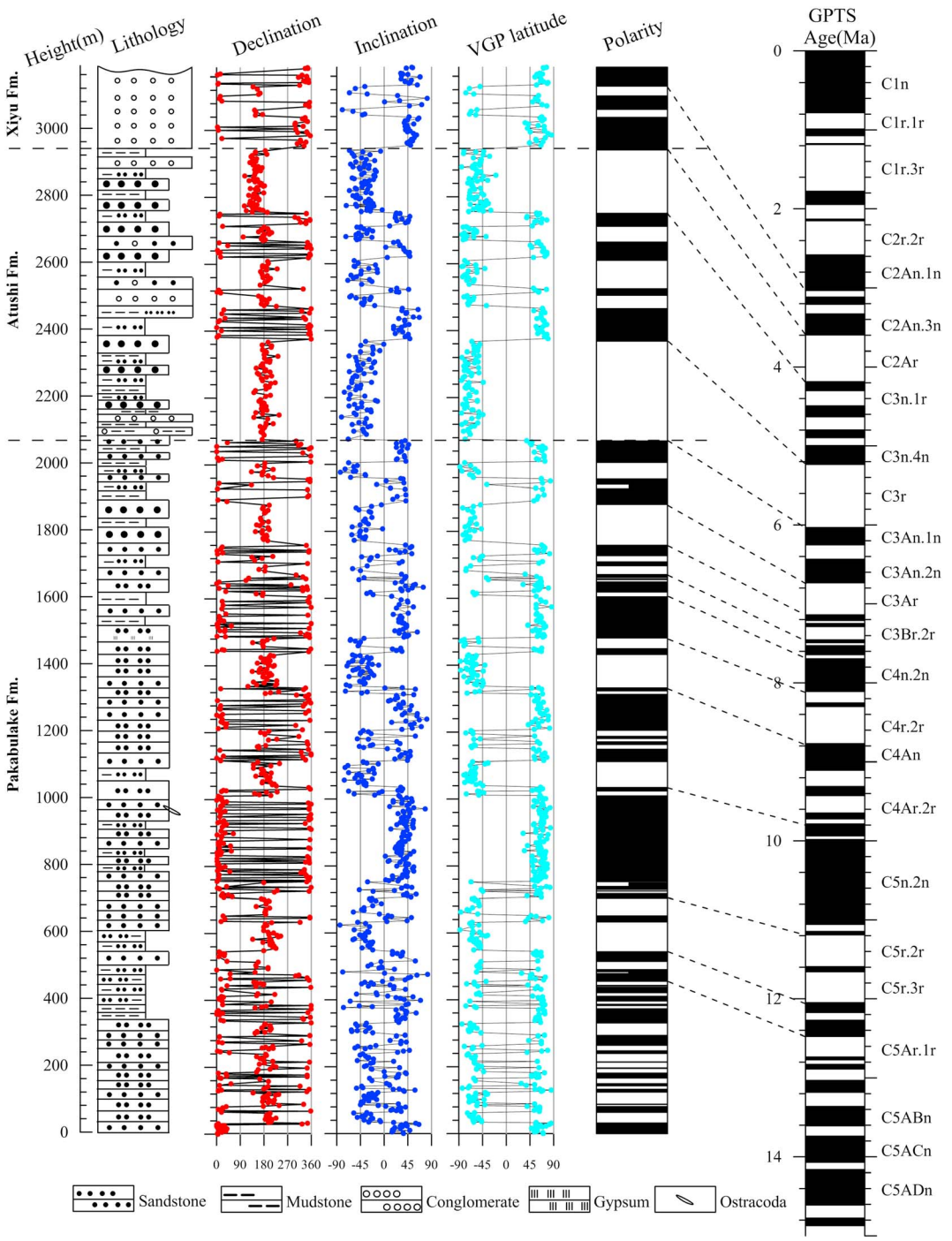


Figure 7. Lithology log and magnetostratigraphic results from the Dashankou section in the Kashi Depression, SW China. The characteristic remanence declination and inclination and VGP latitude are plotted as a function of stratigraphic level and the correlation with the GTS2012 geomagnetic polarity timescale [*Gradstein et al.*, 2012]. Three magnetochrons defined by only one horizon are shown by short half bars.

the critical angle of 4.7° providing an A classification reversal test. In the lower Dashankou section the angular difference between the normal and reversed mean directions is 2.6° with a critical angle of $\gamma_c = 3.4°$ also yielding an A classification reversal test. Hence, the combined evidence from the paleomagnetic reversal and fold tests implies that the ChRM was acquired at, or close to, the time of rock formation and can be confidently interpreted as essentially primary in origin.

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4. Correlation With the GTS2012

Directions of the ChRMs from the 985 horizons have been used to calculate virtual geomagnetic pole (VGP) latitudes and the sampled section organized into stratigraphic levels referred to the VGP latitudes in order to define magnetic polarity zones (Figure 7). The gypsum-bearing Pakabulake Formation is assigned to the Serravallian stage [*Heermance et al.*, 2007] on the basis of the occurrence of a fauna comprising Ostracoda assemblages including *Eucypris insolita* Jiang, *Eucypris* cf. *longa* Mandelstam, *Darwinula ignobilis* Jiang, and *Cyprinotous optalus* Jiang. We therefore consider that deposition of the succession studied here commenced in the mid-Miocene. *Zheng et al.* [2000] find that the Atushi Formation in the Kashi Depression is dominated by a long reversed polarity zone presumed to correlate with the Gilbert reversed chron, while in the Kepintage anticline, *Zhou and Chen* [1990] find assemblages through the correlative stratigraphic interval of the Atushi indicating a Pliocene age; these two lines of evidence provide a chronostratigraphic control for the higher levels of the present study. We therefore correlate the underlying Atushi Formation in the Dashankou section to chrons C3r to C2Ar implying a basal age of ~6.0 Ma (Figure 7). Given the observation that Xiyu conglomerate has generally prograded southward [*Chen et al.*, 2002, 2007; *Heermance et al.*, 2007], we expect the Dashankou section to incorporate a slightly older time interval than the sections studied by *Chen et al.* [2002]. Unfortunately, the lower part of our polarity sequence, which is characterized by a mixed polarity zone below ~500 m, precedes a relatively long reversal zone at 520–680 m and we are unable to correlate this firmly with the GTS2012 [*Gradstein et al.*, 2012]. There are several possible explanations for this including (i) variable rates of sedimentation prevailing during early stages of the infilling of the foreland basin, (ii) possible overlapping or truncation of the section by unidentified faulting, or (iii) an influence of the tectonically induced magnetic fabric recognized by the anisotropy of magnetic susceptibility (AMS) study. We consider that (i) is the only plausible explanation because (ii) is unlikely in the context of careful field observation, while (iii) can be essentially dismissed because the fabric is so weak that it cannot significantly have affected the directional results. Hence, we favor (i) and propose a correlation of this mixed polarity zone with chrons C5Ar.1r to C5ADn of the GTS2012 (Figure 7) to conclude that the Dashankou section was deposited between approximately ~12.4 Ma and ~3.0 Ma (i.e., it is mid-Miocene to late Pliocene in age).

The plot of magnetostratigraphic age versus stratigraphic thickness within the composite section is shown in Figure 8. The single major feature of this plot is an abrupt increase in the sedimentation rate from ~24.8 cm/yr to ~34.5 cm/yr at 6.73 Ma. This corresponds in age to the fourth of the five pulses of sedimentation recognized in the wider studies of the Tianshan noted in section 1. This increase in sediment input in Messinian times also correlates fairly closely with a major sedimentation pulse estimated from the entirely independent method of balanced section reconstruction at ~6–5 Ma in the Tianshan region [*Métivier and Gaudemer*, 1997]. We further note that sedimentation rates inferred here from the magnetostratigraphic correlation are compatible with rates found in foreland basin environments elsewhere which fall in the range 10 to 40 cm/yr [*Burbank et al.*, 1992].

5. Anisotropy of Magnetic Susceptibility (AMS) Results

AMS provides a measure of rock fabric that can potentially result from depositional or postdepositional sedimentary factors, or a combination of both [*Channell et al.*, 1979], or it may alternatively record the imprint of later tectonism. When the relative contributions of these effects can be isolated in the context of the regional geological history, AMS can potentially yield information to help understand the paleogeographic evolution of a sedimentary basin [*Marino and Ellwood*, 1978].

In this study AMS tensors were resolved by application of a KLY-4 Kappabridge susceptibility meter and determined for 985 specimens. The results identify significant changes with lithology and stratigraphic position through the section (Figure 9). Specifically, bulk magnetic susceptibility is lowest (with an average value of 249.9 μSI) near the base of the succession. This is followed by a 3.5 Ma interval (~11–7.5 Ma) of continuing low values below 400 μSI. By ~7.5 Ma (the 1650 m level in the succession) k_m shows significant increase to a relatively higher mean value of 352.8 μSI until ~3.6 Ma (the 2940 m level); k_m increases and remains high thereafter. The overall relatively weak values of bulk susceptibility (mostly < 400 μSI) indicate that anisotropic paramagnetic and antiferromagnetic minerals such as the clays and hematite predominantly control both the susceptibility and anisotropy of these sediments [*Tarling and Hrouda*, 1993]. However, the notable increase of k_m above the ~2940 m level may be due to increased input of small amounts of magnetite after ~3.6 Ma (section 6.1).

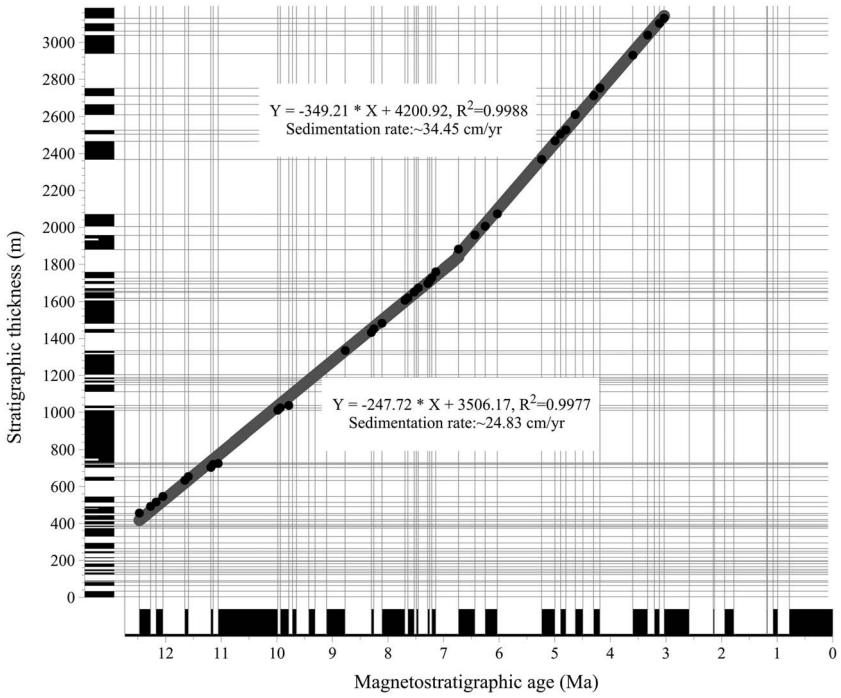


Figure 8. Stratigraphic height as a function of magnetostratigraphic age in the Dashankou section showing accumulation rates in the studied section.

Significant changes in shape parameter (T) and corrected anisotropy degree (P_j) [*Jelinek*, 1981] occur at the 2071 m (6.0 Ma) and 2940 m (3.6 Ma) levels (Figure 9). The former change correlates with an increase in the sedimentation rate as noted in section 4. P_j ranges from 1.002 to 1.396 with a mean of 1.078 and is dependent on lithology. From the base to the 2071 m level in the section, P_j oscillates about a relatively large value (1.1), while marked attenuation of P_j occurs at the 2940 m level. Correlating with this transition, the ellipsoid shapes are dominantly oblate ($T > 0$) from the base up to the 2940 m level; they then change to relatively spherical shapes ($T \approx 0$) in the upper levels of the section (Figure 9).

Stereonet projections of k_{max} and k_{min} principal axes of the AMS ellipsoids are shown in Figure 10. The tilt-corrected k_{max} directions are observed to be well grouped in an approximately ENE-WSW direction with low inclination, and generally parallel to the fold axis, while the tilt-corrected k_{min} directions are oriented predominantly perpendicular to bedding in stratigraphic coordinates (Figures 10a and 10b). The elongated distribution of AMS ellipsoids (P_j-T data) does not pass through, or toward, the magnetite ellipsoid (Figures 10d–10f) and confirms that the observed AMS character is not a straightforward function of magnetite concentration through the succession [*Parés*, 2004].

If the AMS fabrics are of primary sedimentary origin, the k_{max} axial directions would be expected to be distributed predominantly about a N-S direction in conformity with the direction of sediment input and perpendicular to the orogenic front; the k_{min} axes should be roughly perpendicular to bedding but may be expected to record imbrication [e.g., *Hrouda*, 1982; *Tarling and Hrouda*, 1993]. The stereographic projection in Figure 10 of tilt-corrected AMS axial directions shows that k_{min} is indeed approximately perpendicular to bedding as expected but the k_{max} directions tend to be focused along an E-W trend parallel to the axis of the Mutule anticline (Figure 1). This compares with embryonic magnetic fabrics in weakly deformed mudrocks [*Larrasoaña et al.*, 2004; *Parés*, 2004] and suggests that the magnetic fabrics in these rocks record a signature of incipient deformation.

However, a more specific analysis is possible when variations through the succession are considered. When the section is divided into three segments (Figure 10), we observe that samples from the base to the 1880 m level are dominated by oblate fabrics with k_{max} near parallel to the fold axes; the k_{min} axes are broadly grouped around the bedding pole (Figures 10a and 10d). This is likely to have resulted from the imposition

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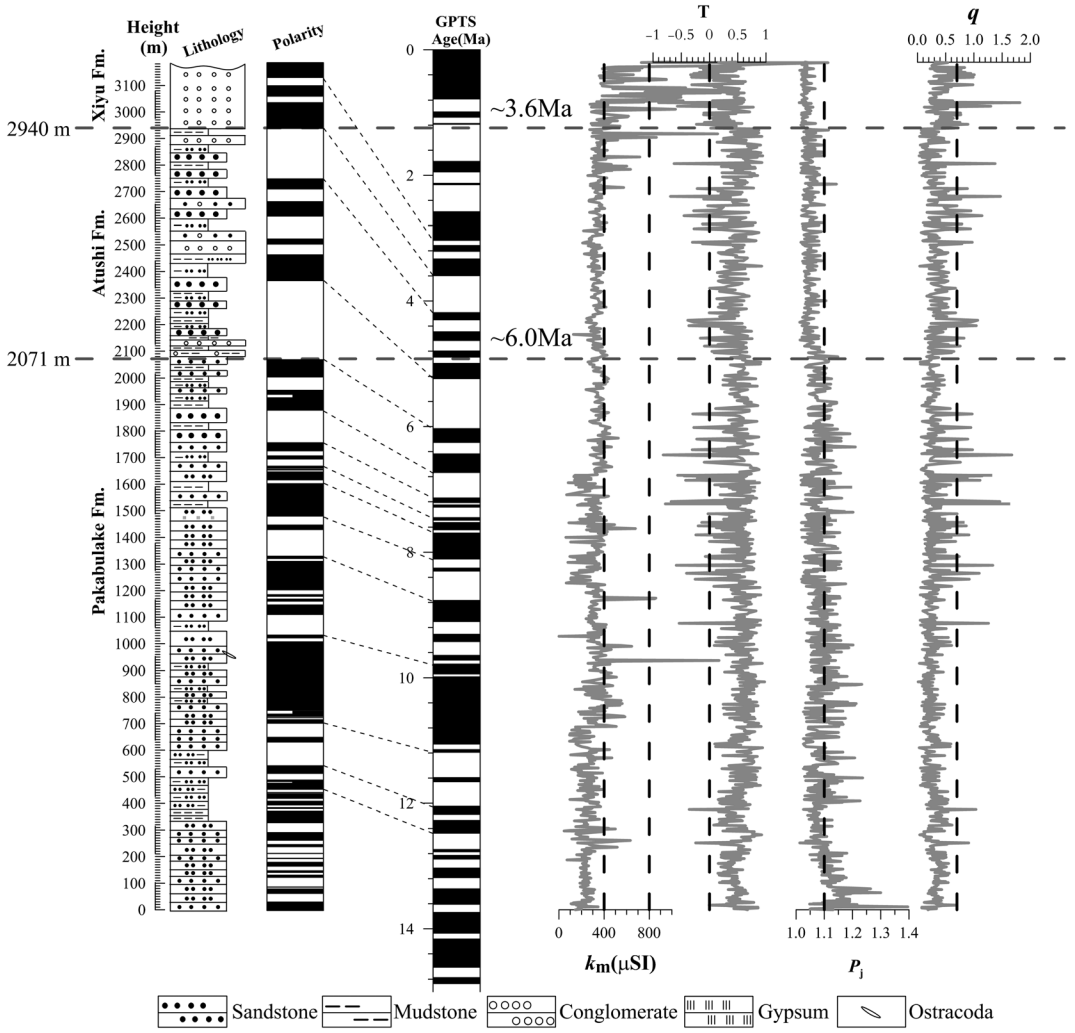


Figure 9. Synthetic lithologic column showing the chronology of the Dashankou section correlated to the GTS2012 timescale with mean magnetic susceptibility (k_m) and fabric properties comprising shape parameter (T), corrected AMS degree (P_j), and magnetic matrix grain degree q of the AMS ellipsoids shown as a function of age.

of tectonic strain and is therefore a signature of the deformation resulting in the open folding. Through the succession between levels 1880 to 2940 m (Figure 10b, 6.7–3.6 Ma), k_{min} axes are clustered around the bedding pole and k_{max} remains parallel to the fold axes but is now more dispersed and the signature of tectonic strain imposition is correspondingly weaker. At higher levels between 2940 and 3180 m there is little or no signature of tectonic strain: The samples show weak and neutral AMS (Figure 10f, 3.6–3.0 Ma), and k_{min} directions are inclined to the northeast (Figure 10c). The Tibetan Plateau and its northern hinterland have been subjected to N-S directed compression since onset of the India-Asia collision, and we suspect that the dominant k_{max} directions observed in the lower part of the succession (Figures 10a and 10b) are a response to compression induced by the regional stress field imparted by this collision. In contrast, above the 2940 m level a signature of tectonic origin is not prominent in the less deeply buried portion of the succession where AMS shapes and k_{min} directions instead show evidence for southward imbrication; they are therefore likely to have a primary sedimentary origin (Figures 10c and 10f).

The AMS T parameter can yield information related to sediment transport conditions when the fabrics are not prominently overprinted by compaction or tectonic-related processes [*Charreau et al.*, 2005; *Gilder et al.*, 2001]. The change in T from oblate before 3.6 Ma to spherical after 3.6 Ma (Figure 9), while reflecting shallower depths of burial and lower levels of straining could also be linked less directly to the tectonism because

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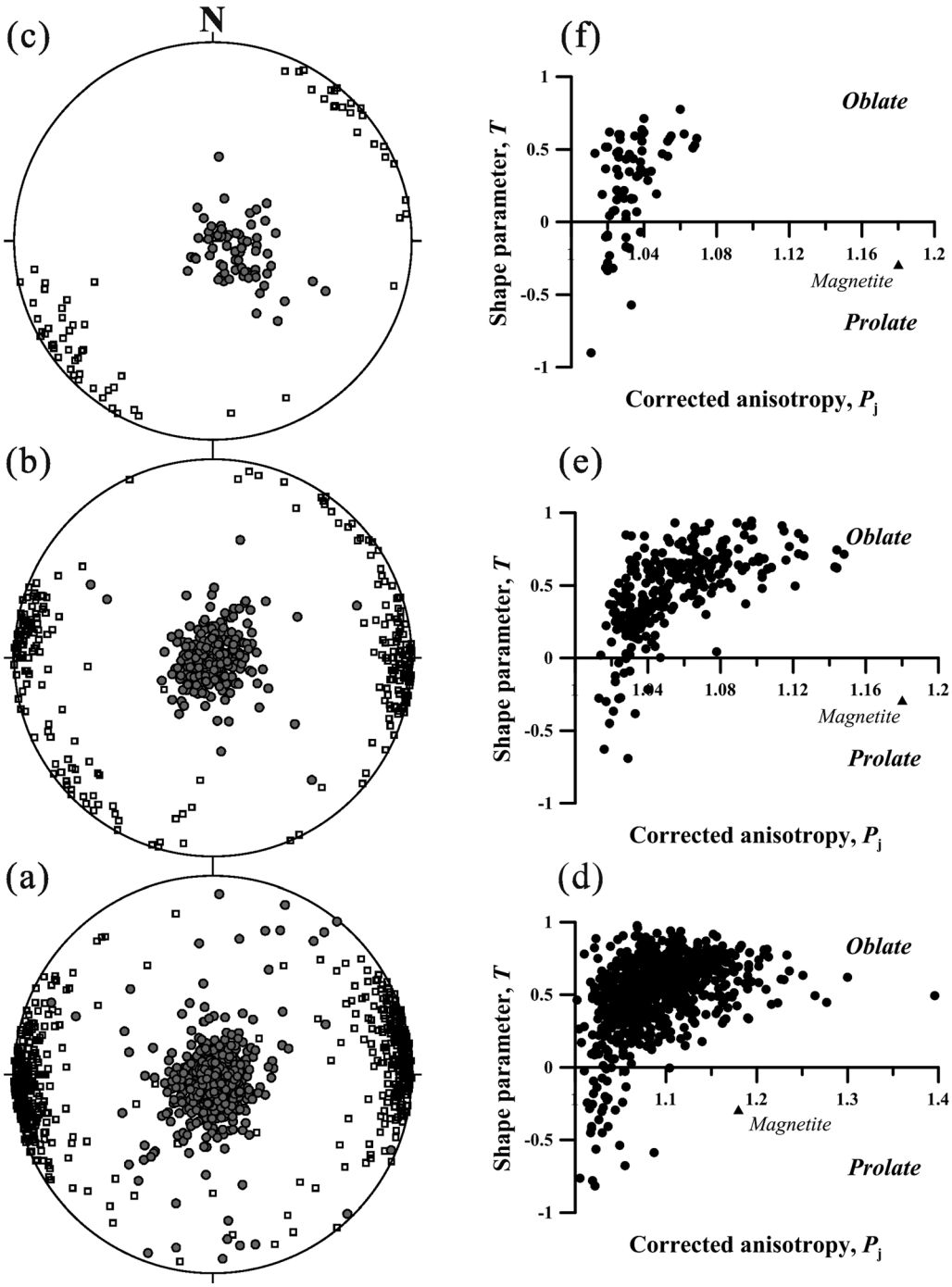


Figure 10. (a–c) Diagrams of AMS principal maximum (square) and minimum (circle) directions in equal-area stereographic projections for the three divisions of the section corresponding to ~11.0–7.5 Ma, ~7.5–3.6 Ma, ~3.6 Ma in Figure 9. (d–f) P_j-T (corrected AMS degree versus AMS shape parameter) diagrams for AMS data for these three divisions.

propagation of thrusting at the margins of the mountain belt would have moved the orogenic front closer to the deposition site and have the effect of reducing the distance traveled by the particles and hence the degree to which they are influenced by abrasion. Thus, in general terms the observed AMS fabrics indicate that the regional stress field following the Cenozoic India-Asia collision is significantly recorded in the lower part of the succession but is much weaker or absent at the higher levels (Figures 9 and 10c).

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6. Discussion

6.1. Basal Age of the Xiyu Conglomerates

The Xiyu conglomerates were first described by Obruchev in the 1930s. Later, T. K. Huang described the lithologies in more detail and assigned the conglomerates to a Xiyu Formation. As originally proposed by *Huang et al.* [1947] in the northern Tianshan, this assignation described a suite of typically massive, 50 to 3200 m thick, molassic deposits comprising dark gray pebble to boulder conglomerates with clast sizes ranging from 3 to >30 cm; minor interbeds of mudstone or sandstone occur where the gradational base defines the regional increase in sedimentation rate. *Huang et al.* [1947] extended this description to cover a coherent and quasi-continuous unit lying at the top of the sedimentary pile in many of the central Asian foreland basins.

There are two opposing explanations for the sedimentary transition defined by the Xiyu Formation, namely, (i) these massive conglomerates are the signature of a late major uplift of the orogeny in response to continuing impingement of India into Asia [*Charreau et al.*, 2009b; *Huang et al.*, 1947, 2010; *Zheng et al.*, 2000] or alternatively, (ii) this coarse clastic input was caused by an increase in erosion rates resulting from climate change [*Burchfiel et al.*, 1999; *Sun et al.*, 2004, 2007; *Zhang et al.*, 2001]. Because the Xiyu Formation cannot be directly dated, the basal ages assigned in the Tianshan region have sometimes reflected prejudices concerning these two possible explanations.

The succession studied here is located in a N-S trending outcrop belt crossing the southern limb of the Kepingtage Anticline and lies ~70 km east of the Boguzihe section studied by *Chen et al*. [2002, Figure 1]. The strata are laterally continuous and can be traced through to the Boguzihe section where the base of the Xiyu Formation is estimated to range from 2.8 Ma on the northwest limb of the Atushi-Talanghe anticline to 1.9 Ma on the southwest limb [*Chen et al.*, 2002, Figure 1]. However, our dating of the magnetostratigraphy in the Dashankou section yields an age of ~3.6 Ma for this boundary and supports the finding of the earlier workers that the Xiyu Formation is diachronous and young toward the south [*Chen et al.*, 2002; *Heermance et al.*, 2007]. This diachronous character of the Xiyu Formation in the Kashi area is therefore likely to have been a consequence of penecontemporaneous uplift resulting in repeated reworking of material degraded from the orogen and consequent progradation of sedimentary input into the foreland basin [*Heermance et al.*, 2007].

However, if the deposition of this formation had been the result of a change in the climatic regime a contemporaneous onset of deposition should be recognized regionally in late Pliocene times at around ~2.6 Ma in NW China [*Sun et al.*, 2004, 2007]. This issue can be further evaluated from the study of rock magnetic parameters because these tend to reflect the changing balance of physical to chemical weathering processes as they respond to degrees of aridification and cooling in the source region [*Balsam et al.*, 2004]. *Huang et al.* [2010] have shown that the climate changes suggested by rock magnetic properties in the Yaha section do indeed correlate with the climate record in the neighboring Kuitun He on the northern piedmont of the Tian Shan [*Sun et al.*, 2007]. Contacts between the Xiyu conglomerates and underlying strata in both the Yaha and Kuitun He sections and the contrasting age estimates for the Xiyu conglomerates show that the essential signature of this formation is independent of wider global climatic changes. Nevertheless, climatic influences will no doubt have modulated local sediment accumulation rates and the impact of the Plio-Pleistocene climatic transition on the Neogene record of orogeny in central Asia remains a topic of continuing debate.

6.2. Implications for the Uplift History of the Southern Tianshan

The Kapingtage (Kalpin) thrust zone is an actively deforming part of the India-Asia collision system and lies south of the Tianshan where it comprises a succession of emergent imbricate structures of arcuate form (Figure 1). This thrust zone is the only margin of the Tarim Basin emplaced as a décollement in the Cenozoic, and both Neogene and Quaternary strata are observed to be folded above the footwall. In the vicinity of our study region these emergent thrust sheets verge mostly toward the interior of the Tarim Basin to the south; they show no major internal deformation and have smoothly sigmoidal profiles with dips ranging typically from ~20° to ~45° [*Allen et al.*, 1999].

The timing of the onset of Cenozoic deformation in the Kepingtage thrust zone is indicated by the change of sediment accumulation rate, a change correlating with a transition from lacustrine to alluvial facies in Messinian times at about 6.7 Ma in the Dashankou section [*Jia et al.*, 2004; *Zheng et al.*, 1999]. As illustrated

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in Figure 8, the increase in sedimentation rate from ~24.8 cm/yr to ~34.5 cm/yr at this time is abrupt and also correlates with input of conglomerates in the section near Dushanzi in the northern Tianshan mountains [*Sun et al.*, 2004]. It is further recognized by a marked increase in sedimentation rate in both the Kuche Depression in the southern Tianshan mountains [*Huang et al.*, 2006; *Jing et al.*, 2011] and in the Jingou river section at the southern margin of the Junggar Basin [*Ji et al.*, 2008].

Because the Tarim Basin is enclosed, the sediment accumulation rates are controlled primarily by rates of sediment supply and the accommodation space. Sediment supply is a function of climate [*Molnar*, 2004], source rock lithology [*Carroll et al.*, 2006] and proximity to the sediment source [*Burbank and Beck*, 1991]. The accommodation space is created either by tectonically induced subsidence beneath a faulted block [*Beaumont*, 1981] or by uplift that serves to trap sedimentation within a Piggyback Basin [*DeCelles and Giles*, 1996]. Thus, an abrupt change in the sediment accumulation rate could be the result of either a change in sediment flux [*Molnar*, 2001; *Zhang et al.*, 2001] or tectonic subsidence [*Paola et al.*, 1992] or by a combination of both. Balanced section restoration of compressional deformation in the Kalpin Uplift has yielded an estimate for a minimum shortening of ~22–28% during the late Cenozoic [*Allen et al.*, 1999; *Yin et al.*, 1998], based on the field observations and reconstructed cross sections. We therefore interpret the change in sedimentation rate at 6.7 Ma as a tectonically influenced phenomena linked to the Kalpin Uplift.

Moreover, the dramatic facies and sedimentation rate changes at this time could also reflect thrust loading in the southern Tianshan. Thus, as pointed out by *Jordan et al.* [1988], an increase in the input of sediments to a foreland basin may or may not be related to thrusting, depending on whether climate conditions were changing to increase or decrease the power of erosion. However, the relative contributions of tectonics and climate to the erosion processes are still a subject of controversy [*Burbank et al.*, 2003, 1993; *Molnar and England*, 1990; *Sun et al.*, 2004]. Climate records in both west Antarctica [*Kennett and Barker*, 1990] and the Arctic [*Thiede and Vorren*, 1994] indicate cooling and ice sheet expansion after 6 Ma, and stable carbon isotopic studies of fossil soils from Pakistan and North America reveal an expansion of C_4 grasses at 8–6 Ma indicating the development of an arid climate commencing in the late Miocene [*Cerling et al.*, 1993]. Given that the studied section is located in a region of active tectonics, the increase in abundance of the arid species of *Artemisia* and *Chenopodiaceae* after 6 Ma in the Taxihe section [*Sun and Zhang*, 2008] may have resulted from uplift of the mountain range causing it to act as a barrier to the influence of the Indian and possibly also the Atlantic Oceans on the climate of the continental interior. Although climatic changes no doubt occurred during deposition of the Dashankou section, we consider that the facies change is primarily a response to late tectonic reactivation in the Tianshan.

7. Conclusion

The detailed paleomagnetic study of the Dashankou section has provided a high-resolution chronology for the Neogene sequence in the Kashi Depression in the southwest part of the Tianshan Mountains. On the basis of the chrondurations and Ostracoda assemblages found in this section, our preferred magnetostratigraphic correlation suggests that the sampled 3187 m thick section was deposited between ~12.4 Ma and ~3.0 Ma, i.e., middle Miocene (Serravallian) and late Pliocene times. A rapid increase in sedimentation rate at ~6.7 Ma is interpreted to record onset of rejuvenated uplift of the Tianshan Mountains. While climatic changes may have modulated the sedimentary record, they do not appear to be a dominant control on sediment accumulation between 7.0 and 2.58 Ma. Magnetic fabrics identify the influence of the regional stress field imparted by ongoing India-Asia collision in the lower part of the succession contrasting with a dominant sedimentary fabric preserved in the higher parts of the succession. Together with other magnetostratigraphic studies conducted around the Tianshan, our study confirms that onset of coarse rudaceous deposition defined as the Xiyu Formation is diachronous and progrades southward into the foreland basin. Hence, this formation cannot be considered to be a chronostratigraphic marker related to any particular tectonic or climatic event.

Acknowledgments

The data for this paper are available as in the supporting information Tables S1–S3. This study was financed by 100 Talents Program of Xinjiang and West Light Foundation of CAS, Innovation Talents Project of Xinjiang, and the National Nature Science Foundation of China (grant 40525013 and 41404054). We are grateful to Sun D.J., Yi Z.Y., Chen L.W., and Tang X.D. for their contributions to the field sampling and constructive discussions.

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